What is the effect of ancestral state estimation on disparity?

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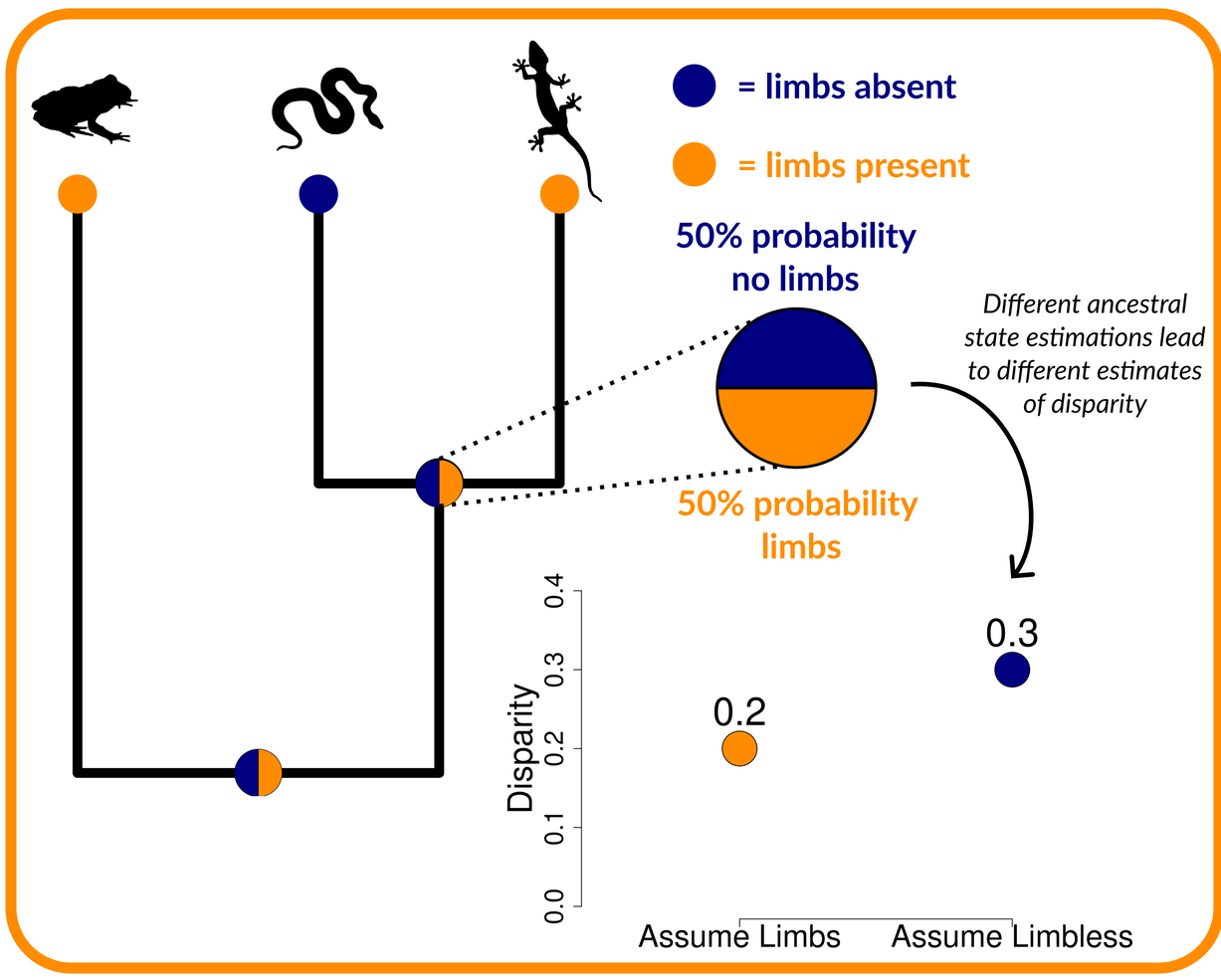
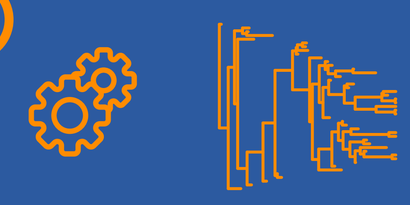


1 Tree simulations



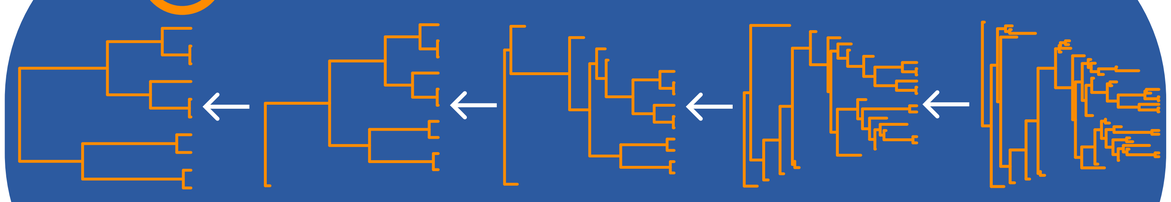
Simulated 100 trees using birth-death algorithm.

2 Trait simulations

Fast | Medium | Slow

Discrete trait data simulated using three different transition rates across each tree.

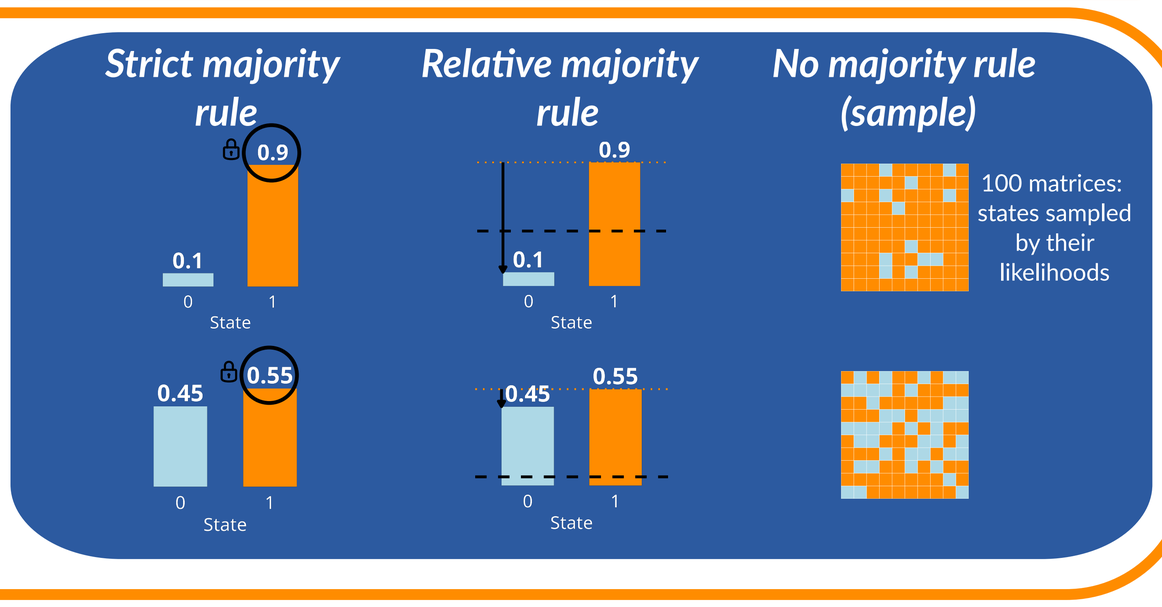
3 Fossil sampling



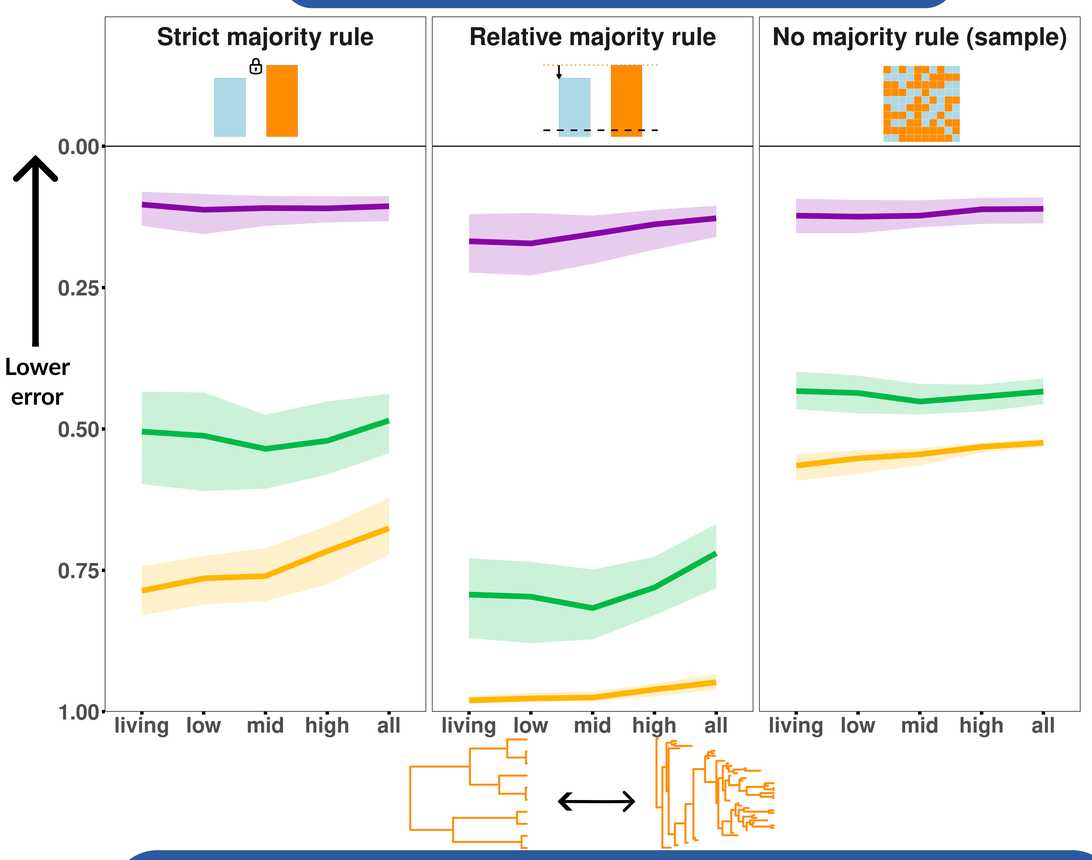
Prune each tree and matrix across a range of intensities to simulate different fossil sampling levels

4 Ancestral state estimation

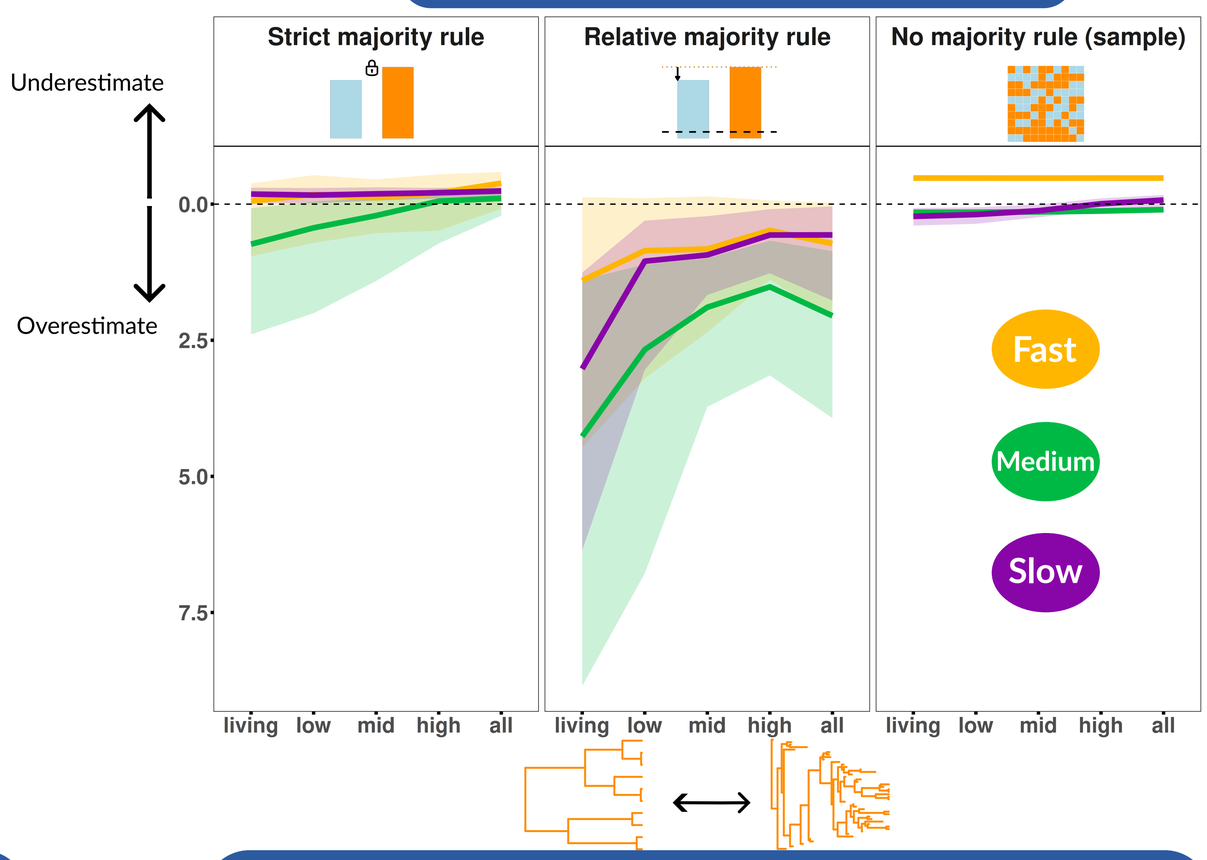
- Maximum Likelihood estimates the probability of each possible state at a node.
- A dominant state is simple to infer — but when states are equally likely, we face uncertainty.
- We use three majority rules to handle uncertainty:
  - Strict majority rule = **permissive**
  - Relative majority rule = **conservative**
  - No majority rule = **propagates uncertainty**



Ancestral state estimation error



Disparity error



Conclusions

- **No majority rule returns lowest error** for both raw state and disparity estimates.
- **True disparity value often recovered** even when raw ancestral state estimation accuracy is low.
- Highlights importance of **propagating uncertainty** into macroevolutionary workflows.

Next steps

- Investigate how **continuous characters** compare to discrete state estimation and disparity.
- Benchmark findings with **empirical datasets** of exceptional fossil sampling, e.g., Foraminifera.
- Explore how **mass extinction events (random and selective)** can influence ancestral state estimation.